Consider the examples,

$$\begin{array}{r|l} & 0.875 \\ 8 & 7.000 \\ & 64 \\ \hline & 60 \\ & 56 \\ \hline & 40 \\ & 40 \\ \hline & 0 \end{array}$$

$\frac{7}{8} = 0.875$

$$\begin{array}{r|l} & 0.84 \\ 25 & 21.00 \\ & 200 \\ \hline & 100 \\ & 100 \\ \hline & 0 \end{array}$$

$\frac{21}{25} = 0.84$

$$\begin{array}{r|l} & 2.71875 \\ 32 & 87.00000 \\ & 64 \\ \hline & 230 \\ & 224 \\ \hline & 60 \\ & 32 \\ \hline & 280 \\ & 256 \\ \hline & 240 \\ & 224 \\ \hline & 160 \\ & 160 \\ \hline & 0 \end{array}$$

$\frac{-87}{32} = -2.71875$

Note

These show that the process could lead to a decimal with finite number of decimal places. They are called terminating decimals.

Can the decimal representation of a rational number lead to forms of decimals that do not terminate? The following examples (with non-zero remainder) throw some light on this point.

Example 2.2 Represent the following as decimal form (i) $\frac{-4}{11}$ (ii) $\frac{11}{75}$

Solution

$$\begin{array}{r|l} & 0.3636\ldots. \\ 11 & 4.0000 \\ & 33 \\ \hline & 70 \\ & 66 \\ \hline & 40 \\ & 33 \\ \hline & 70 \\ & 66 \\ \hline & 4 \\ & \vdots \end{array}$$

$$\begin{array}{r|l} & 0.1466\ldots \\ 75 & 11.0000 \\ & 7\,5 \\ \hline & 350 \\ & 300 \\ \hline & 500 \\ & 450 \\ \hline & 500 \\ & 450 \\ \hline & 50 \\ & \vdots \end{array}$$

Thus we see that, $\frac{-4}{11} = -0.\overline{36}$ $\frac{11}{75} = 0.14\overline{6}$

The reciprocals of Natural Numbers are Rational numbers. It is interesting to note their decimal forms. See the first ten.

S.No.	Reciprocal	Decimal Representation
1	$\frac{1}{1} = 1.0$	Terminating
2	$\frac{1}{2} = 0.5$	Terminating
3	$\frac{1}{3} = 0.\overline{3}$	Non-terminating Recurring
4	$\frac{1}{4} = 0.25$	Terminating
5	$\frac{1}{5} = 0.2$	Terminating
6	$\frac{1}{6} = 0.1\overline{6}$	Non-terminating Recurring

2-Real Numbers.indd 47 02/12/2022 21:33:14